General Mathematics (Further Maths 2020–22 set) · Harder paper

Real VCAA questions ranked by the published state results. Editable: type your answers straight into this PDF, save it, then mark yourself with the guide at the back.

Name _____ **Date** _____

Exam 1 · multiple choice	Exam 2 · written response	State average on these questions
40 marks	**58 marks**	**39 / 98 (40%)**
reading 15 min · writing 1 h 30 min	reading 15 min · writing 1 h 30 min	a student at the state average on every question

How to use this paper

1. Open each question with its blue link. It opens the official VCAA exam PDF at the right page. The question text isn't reproduced here because it is VCAA copyright.
2. Answer in this file: click a bubble for multiple choice, and type in the boxes for written answers (long answers scroll inside the box). Working can be typed in plain text, e.g. 0.004 × 5440.88 = 21.76.
3. Keep to the real time limits above, with your CAS, your bound reference and the VCAA formula sheet, and don't look at the guide until you've finished.
4. Save with Ctrl/Cmd + S. Use a PDF app that supports forms: Adobe Acrobat Reader, Chrome, Edge, Firefox, macOS Preview or Xodo.
5. Mark yourself: in the guide at the back, tick each marking point you earned, enter your marks, and compare your total with the state average.

Why these questions

These are real VCAA questions from the 2020–2022 Further Mathematics exams, the subject General Maths replaced in 2023. Only the core (data analysis, recursion and finance) and the Matrices and Networks modules are used, and questions on content that isn't in the current General Maths study design (solving simultaneous equations with matrices) are left out. They were ranked by how Further Maths students actually scored on them within each area of study, and this paper takes the lowest band.

Sources

- 2020: Exam 1 (PDF) · report · Exam 2 (PDF) · report
- 2021: Exam 1 (PDF) · report · Exam 2 (PDF) · report
- 2022: Exam 1 (PDF) · report · Exam 2 (PDF) · report

Exam 1 · multiple choice (40 marks, reading 15 min, writing 1 h 30 min)

Open each question in the official exam PDF, then click your answer. Further Maths questions have five options (A–E).

Data analysis

Question 1 · 1 mark

2020 VCAA Further Maths Exam 1 · Core · Question 4 · open in the official exam PDF (p. 5) ↗

Which value could be the third quartile of 252 forearm circumferences shown in a histogram

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 2 · 1 mark

2020 VCAA Further Maths Exam 1 · Core · Question 13 · open in the official exam PDF (p. 9) ↗

What is always true of a least squares line

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 3 · 1 mark

2020 VCAA Further Maths Exam 1 · Core · Question 15 · open in the official exam PDF (p. 10) ↗

How March rainfall (seasonal index 0.741) must change to be deseasonalised

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 4 · 1 mark

2020 VCAA Further Maths Exam 1 · Core · Question 19 · open in the official exam PDF (p. 12) ↗

Median monthly airline passengers in 1960, from a time series plot

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 5 · 1 mark

2021 VCAA Further Maths Exam 1 · Core · Question 3 · open in the official exam PDF (p. 6) ↗

Which two-way frequency table could match the segmented bar chart

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 6 · 1 mark

2021 VCAA Further Maths Exam 1 · Core · Question 6 · open in the official exam PDF (p. 8) ↗

Best display for resting pulse rate against age group

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 7 · 1 mark

2021 VCAA Further Maths Exam 1 · Core · Question 7 · open in the official exam PDF (p. 8) ↗

Number of 800 auditionees who scored below 76.0 (mean 69.5, SD 6.5)

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 8 · 1 mark

2021 VCAA Further Maths Exam 1 · Core · Question 11 · open in the official exam PDF (p. 10) ↗

How many of 10 adults have a predicted weight above their actual weight

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 9 · 1 mark

2021 VCAA Further Maths Exam 1 · Core · Question 12 · open in the official exam PDF (p. 11) ↗

Best description of a quarterly sales time series over 11 years

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 10 · 1 mark

2021 VCAA Further Maths Exam 1 · Core · Question 13 · open in the official exam PDF (p. 12) ↗

Nine-median smoothed points for game 10, from a time series plot

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 11 · 1 mark

2021 VCAA Further Maths Exam 1 · Core · Question 14 · open in the official exam PDF (p. 12) ↗

Three-mean smoothed value for Thursday, given a five-mean value and some daily figures

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 12 · 1 mark

2021 VCAA Further Maths Exam 1 · Core · Question 16 · open in the official exam PDF (p. 13) ↗

Predicted actual visitors for February from a deseasonalised trend line and seasonal indices

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 13 · 1 mark

2022 VCAA Further Maths Exam 1 · Core · Question 1 · open in the official exam PDF (p. 4) ↗

Shape of a histogram of possum skull widths

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 14 · 1 mark

2022 VCAA Further Maths Exam 1 · Core · Question 13 · open in the official exam PDF (p. 10) ↗

Median head length of 17 crocodiles, read from the scatterplot

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 15 · 1 mark

2022 VCAA Further Maths Exam 1 · Core · Question 14 · open in the official exam PDF (p. 11) ↗

Percentage of variation in body length not explained by head length, given r = 0.963

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 16 · 1 mark

2022 VCAA Further Maths Exam 1 · Core · Question 16 · open in the official exam PDF (p. 11) ↗

How summer sales with seasonal index 1.25 must change to be deseasonalised

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Recursion and financial modelling

Question 17 · 1 mark

2020 VCAA Further Maths Exam 1 · Core · Question 26 · open in the official exam PDF (p. 14) ↗

Rule for the balance after n years at 3% p.a. compounding quarterly

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 18 · 1 mark

2020 VCAA Further Maths Exam 1 · Core · Question 30 · open in the official exam PDF (p. 15) ↗

Original sum invested, given two interest rates over 20 years and a $10 000 top-up halfway

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 19 · 1 mark

2021 VCAA Further Maths Exam 1 · Core · Question 21 · open in the official exam PDF (p. 15) ↗

Effective annual rate, given $3000 grows to $3728.92 in four years compounding monthly

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 20 · 1 mark

2021 VCAA Further Maths Exam 1 · Core · Question 23 · open in the official exam PDF (p. 15) ↗

Value of R in a monthly loan recurrence relation, given the loan is repaid in 20 years

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 21 · 1 mark

2021 VCAA Further Maths Exam 1 · Core · Question 24 · open in the official exam PDF (p. 16) ↗

New interest rate that lets an interest-only start still repay the loan in 20 years

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 22 · 1 mark

2022 VCAA Further Maths Exam 1 · Core · Question 17 · open in the official exam PDF (p. 12) ↗

R_2 from $R_0 = 2$, $R_{n+1} = 2 - R_n$

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 23 · 1 mark

2022 VCAA Further Maths Exam 1 · Core · Question 21 · open in the official exam PDF (p. 13) ↗

How many of four statements about nominal and effective rates are true

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 24 · 1 mark

2022 VCAA Further Maths Exam 1 · Core · Question 24 · open in the official exam PDF (p. 14) ↗

Quarterly addition C, given the balance after two years of 0.52% per quarter

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Matrices

Question 25 · 1 mark

2020 VCAA Further Maths Exam 1 · Matrices module · Question 7 · open in the official exam PDF (p. 22) ↗

Percentage buying at Giorgio's on Friday, from Monday's split and a transition matrix

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 26 · 1 mark

2020 VCAA Further Maths Exam 1 · Matrices module · Question 8 · open in the official exam PDF (p. 22) ↗

Order of the product $C^T \times (A^T \times B)^T$ from the orders of A, B and C

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 27 · 1 mark

2020 VCAA Further Maths Exam 1 · Matrices module · Question 10 · open in the official exam PDF (p. 23) ↗

Which statement about the unknown entries of a transition matrix is true, given S_0 and S_1

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 28 · 1 mark

2021 VCAA Further Maths Exam 1 · Matrices module · Question 5 · open in the official exam PDF (p. 20) ↗

Which matrix expression is defined, given A is 7 × 7 and B is 10 × 7

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 29 · 1 mark

2021 VCAA Further Maths Exam 1 · Matrices module · Question 7 · open in the official exam PDF (p. 21) ↗

S_2 from $S_{n+1} = TS_n - C$, given T, S_0 and S_1

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 30 · 1 mark

2021 VCAA Further Maths Exam 1 · Matrices module · Question 8 · open in the official exam PDF (p. 21) ↗

Percentage of marsupials expected to stay at the same feeding station from Tuesday to Wednesday

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 31 · 1 mark

2022 VCAA Further Maths Exam 1 · Matrices module · Question 7 · open in the official exam PDF (p. 19) ↗

Where element m_{31} goes when a column matrix is multiplied by a permutation matrix

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 32 · 1 mark

2022 VCAA Further Maths Exam 1 · Matrices module · Question 8 · open in the official exam PDF (p. 20) ↗

Matrix calculation for each person's total time to service every department's computers

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Networks and decision mathematics

Question 33 · 1 mark

2020 VCAA Further Maths Exam 1 · Networks module · Question 7 · open in the official exam PDF (p. 27) ↗

How many of four statements hold for a graph of people and the ice-cream flavours they chose

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 34 · 1 mark

2020 VCAA Further Maths Exam 1 · Networks module · Question 10 · open in the official exam PDF (p. 30) ↗

Effect of moving H's workers to F (F drops to 3 weeks, but F and H can't run together)

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 35 · 1 mark

2021 VCAA Further Maths Exam 1 · Networks module · Question 3 · open in the official exam PDF (p. 23) ↗

Number of faces of a graph drawn with crossing edges

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 36 · 1 mark

2021 VCAA Further Maths Exam 1 · Networks module · Question 5 · open in the official exam PDF (p. 24) ↗

How many of five statements (planar, cycle, bridge, Eulerian trail, Hamiltonian path) hold for a graph

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 37 · 1 mark

2021 VCAA Further Maths Exam 1 · Networks module · Question 7 · open in the official exam PDF (p. 25) ↗

Number of zeros in the adjacency matrix of a network with a loop and multiple paths

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 38 · 1 mark

2022 VCAA Further Maths Exam 1 · Networks module · Question 3 · open in the official exam PDF (p. 22) ↗

Allocation of four athletes to four events that maximises the total distance

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 39 · 1 mark

2022 VCAA Further Maths Exam 1 · Networks module · Question 4 · open in the official exam PDF (p. 23) ↗

Number of edges to remove so a graph with crossings becomes planar

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 40 · 1 mark

2022 VCAA Further Maths Exam 1 · Networks module · Question 8 · open in the official exam PDF (p. 24) ↗

Sum of all float times for the same project

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Exam 2 · written response (58 marks, reading 15 min, writing 1 h 30 min)

Open each question in the official exam PDF, then type your answer to every part in the boxes. Show working where more than one mark is available.

Data analysis

Question 1 (7 marks)

2020 VCAA Further Maths Exam 2 · Core · Question 5 · open in the official exam PDF (p. 10) ↗

Body density against waist measurement for 250 men: drawing, using and checking a given least squares line

a. Draw the least squares line on the scatterplot (1 mark)

b. Predicted body density for a 65 cm waist, to two decimal places (1 mark)

c. Is that prediction extrapolation or interpolation? (1 mark)

d. Interpret the slope (1 mark)

e. Show that the residual for a 122 cm waist and density 0.995 is −0.02 (1 mark)

f. Correlation coefficient from $r^2 = 0.6783$, to three decimal places (1 mark)

g. Does the residual plot support linearity? Explain (1 mark)

Question 2 (5 marks)

2020 VCAA Further Maths Exam 2 · Core · Question 6 · open in the official exam PDF (p. 12) ↗

Mean height against mean age for seven age groups of women: a line through two points, then a transformation to linearise

a. Difference between the mean heights of women in their twenties and their eighties (1 mark)

b. Describe the association: strength and direction (1 mark)

c. Equation of the line through (20, 168) and (85, 157), to three significant figures (1 mark)

d. Transform mean age to linearise the data and fit a least squares line, to four significant figures (2 marks)

Question 3 (7 marks)

2021 VCAA Further Maths Exam 2 · Core · Question 1 · open in the official exam PDF (p. 2) ↗

High jump, shot-put and javelin results for 15 heptathletes: variables, mean, z-scores, the normal model and boxplots

a. Number of numerical variables in the table (1 mark)

b. Mean high jump height (1 mark)

c. Standardised score for a 14.50 m shot-put throw (mean 13.74, SD 1.43), to one decimal place (1 mark)

d. Percentage expected to jump higher than an athlete with z = −1.0 (1 mark)

e. Why the high jump boxplot has no upper whisker (1 mark)

f. Minimum javelin throw that would show as an upper outlier (2 marks)

Question 4 (5 marks)

2021 VCAA Further Maths Exam 2 · Core · Question 4 · open in the official exam PDF (p. 8) ↗

Men's and women's 100 m freestyle winning times since 1912, each with a least squares trend line

a. Draw the women's least squares line on the plot (1 mark)

b. Predicted women's minus men's winning time for 2024, to one decimal place (2 marks)

c. First Olympic year the lines predict women will be faster than men (2 marks)

Recursion and financial modelling

Question 5 (2 marks)

2020 VCAA Further Maths Exam 2 · Core · Question 11 · open in the official exam PDF (p. 18) ↗

A loan at 4.1% p.a. with known balances after four and seven years: how many more repayments are needed

Number of further monthly repayments after the first seven years (2 marks)

Question 6 (3 marks)

2021 VCAA Further Maths Exam 2 · Core · Question 8 · open in the official exam PDF (p. 13) ↗

A \$570 000 loan with fortnightly interest: $S_0 = 570\,000$, $S_{n+1} = 1.001S_n - 1193$

a. Balance after the first repayment (1 mark)

b. Show the interest rate is 2.6% p.a. (1 mark)

c. Final (larger) repayment that clears the loan, to the nearest cent (1 mark)

Question 7 (3 marks)

2021 VCAA Further Maths Exam 2 · Core · Question 9 · open in the official exam PDF (p. 14) ↗

A \$152 431 annuity paying \$900 a month for 25 years at 5.1% p.a., then a rate cut after two years

a. Value after two months, showing recursive calculations (2 marks)

b. One-off amount to add after two years so the payments last the same time at 4.6% p.a. (1 mark)

Question 8 (4 marks)

2022 VCAA Further Maths Exam 2 · Core · Question 8 · open in the official exam PDF (p. 14) ↗

A \$580 000 loan modelled by $L_0 = 580\,000$, $L_{n+1} = 1.002L_n - 3045.26$

a. Balance after two months, showing recursive calculations (1 mark)

b. Annual interest rate (1 mark)

c. Amount by which the final repayment is smaller, to the nearest cent (1 mark)

d. Multiplying factor that would repay the loan one year sooner, to four decimal places (1 mark)

Matrices

Question 9 (5 marks)

2021 VCAA Further Maths Exam 2 · Matrices module · Question 3 · open in the official exam PDF (p. 18) ↗

Monthly switching between three olive oil brands for 8000 shoppers, then a model with extra shoppers added each month

a. How many shoppers bought a different brand in August from July (1 mark)

b. Complete S_1 (1 mark)

c. Percentage of August's Carmani buyers who also bought Carmani in July (1 mark)

d. Calculation showing Ohana is bought by 50% of shoppers in the long run (1 mark)

e. Value of k in $R_{n+1} = TR_n + B$, given R_2 (1 mark)

Question 10 (2 marks)

2021 VCAA Further Maths Exam 2 · Matrices module · Question 4 · open in the official exam PDF (p. 20) ↗

A round-robin video game tournament: complete a results matrix from one- and two-step dominance totals

Complete the results matrix using all the information (2 marks)

Question 11 (3 marks)

2022 VCAA Further Maths Exam 2 · Matrices module · Question 4 · open in the official exam PDF (p. 20) ↗

Votes for three ski captain candidates changing day to day, then a candidate withdraws

a. Number planning to vote for Ali after one day (1 mark)

b. Who becomes captain after Max withdraws, and with how many votes (2 marks)

Networks and decision mathematics

Question 12 (4 marks)

2020 VCAA Further Maths Exam 2 · Networks module · Question 3 · open in the official exam PDF (p. 28) ↗

Three training routes through a weighted graph of exercise stations, all starting at S

a. Shortest distance from S to O (1 mark)

b.i. Name for a route along every track exactly once (1 mark)

b.ii. Station where that route finishes (1 mark)

c. Track that must be repeated to visit every station and return to S in the minimum distance (1 mark)

Question 13 (4 marks)

2020 VCAA Further Maths Exam 2 · Networks module · Question 5 · open in the official exam PDF (p. 31) ↗

A nine-activity building project given as a table of EST, durations and predecessors, with one duration missing

a. Where the dummy activity goes (1 mark)

b. Duration of activity C (1 mark)

c. The four activities that have float (1 mark)

d. Minimum completion time if one activity is crashed (1 mark)

Question 14 (4 marks)

2022 VCAA Further Maths Exam 2 · Networks module · Question 3 · open in the official exam PDF (p. 25) ↗

A directed network of stormwater pipes with capacities in litres per minute

a. Capacity of Cut 1 (1 mark)

b. Maximum flow from source to sink (1 mark)

c. Pipe to reverse for the largest increase in flow (1 mark)

d. Pipe whose capacity should be increased for the largest gain, and its new capacity (1 mark)

Marking guide

Mark yourself only after you've finished both exams. Tick the boxes you earned, enter your marks and compare with how Victoria went on each question.

Exam 1 · multiple choice

✓	Q	Ans	Why	How Victoria went
☐	1	**C**	A quarter of 252 is 63. Counting down from the top, the bars above 30.5 cm hold fewer than 63 men and adding the 30.0–30.5 cm bar passes 63, so Q3 lies in that class: 30.3.	50% correct A 7% · B 13% · C 50% · D 24% · E 5%
☐	2	**E**	It minimises the sum of the squared vertical distances (residuals) from the points to the line. Only a third of students knew this.	34% correct A 11% · B 23% · C 21% · D 10% · E 34%
☐	3	**D**	Deseasonalised = actual ÷ 0.741 = 1.35 × actual: increase by 35%. A and B used 1 − 0.741.	43% correct A 17% · B 24% · C 9% · D 43% · E 6%
☐	4	**A**	Put the 12 monthly values in order: the 6th and 7th are both about 461 000, so that is the median. D came from using the middle months of the year instead of the middle values.	45% correct A 45% · B 7% · C 9% · D 32% · E 6%
☐	5	**A**	Work out column percentages: in A, 91 ÷ 140 = 65% and 90 ÷ 200 = 45% domestic, matching the chart. C reads the percentages straight in as counts with the rows swapped.	50% correct A 50% · B 13% · C 28% · D 1% · E 8%
☐	6	**C**	Pulse rate is numerical and age group is categorical with four groups: parallel boxplots. A back-to-back stem plot only handles two groups.	51% correct A 28% · B 13% · C 51% · D 3% · E 4%
☐	7	**D**	76 is one SD above the mean, so 84% score below it: 0.84 × 800 = 672. B counted the successful ones.	54% correct A 2% · B 32% · C 9% · D 54% · E 3%
☐	8	**D**	Fit weight against height (height is explanatory) and look at the residuals: six are negative, so the prediction is too high six times.	40% correct A 14% · B 29% · C 11% · D 40% · E 5%
☐	9	**D**	Sales drift down over time with irregular ups and downs, but the peaks don't recur at the same quarter each year: a decreasing trend with irregular fluctuations. More students wrongly saw seasonality (E) than chose the right answer.	34% correct A 4% · B 8% · C 13% · D 34% · E 41%
☐	10	**C**	Take games 6 to 14 and find the middle (5th) value of those nine scores: 110.	53% correct A 4% · B 8% · C 53% · D 24% · E 11%
☐	11	**B**	Five-mean for Thursday = 206, so Tuesday to Saturday total 1030. Take off Tuesday (186) and Saturday (346): Wednesday to Friday total 498, so the three-mean is 498 ÷ 3 = 166.	48% correct A 8% · B 48% · C 15% · D 25% · E 3%
☐	12	**E**	Deseasonalised prediction for month 2 = 2349 − 198.5 × 2 = 1952. Re-seasonalise: 1952 × 1.25 = 2440. C forgot to re-seasonalise.	52% correct A 7% · B 4% · C 29% · D 7% · E 52%
☐	13	**C**	Apart from one isolated value at 67–68 mm, the longer tail runs to the left (down to 49 mm), so the distribution is negatively skewed with a possible outlier. Most students called it symmetric (E).	36% correct A 1% · B 1% · C 36% · D 8% · E 54%
☐	14	**B**	Order the 17 head lengths: the 9th is about 51 cm. D and E used body length.	51% correct A 8% · B 51% · C 20% · D 14% · E 7%

☐	15	**C**	$r^2 = 0.963^2 \approx 0.927$, so 92.7% is explained and 7.3% is not.	48% correct A 7% · B 24% · C 48% · D 13% · E 7%
☐	16	**A**	Deseasonalised = actual ÷ 1.25 = 0.8 × actual: reduce by 20%. B (25%) is a common trap.	40% correct A 40% · B 24% · C 6% · D 7% · E 22%
☐	17	**E**	3% p.a. is 0.75% per quarter and there are 4n quarters in n years: $R_n = 5000 \times 1.0075^{4n}$. D counts quarters, not years.	20% correct A 5% · B 18% · C 6% · D 51% · E 20%
☐	18	**B**	Work back: $686\,904.09 \div (1 + 0.028/12)^{120} - 10\,000$, then $\div (1 + 0.032/12)^{120} \approx$ \$370 000. Finance Solver in two steps does the same.	41% correct A 10% · B 41% · C 18% · D 15% · E 15%
☐	19	**D**	Monthly growth factor = (3728.92 ÷ 3000)^(1/48); the effective rate is that factor to the power 12, minus 1 ≈ 5.59%. A is the nominal rate.	40% correct A 22% · B 12% · C 13% · D 40% · E 12%
☐	20	**A**	Finance Solver with N = 240, PV = 450 000, PMT = −2633, FV = 0 gives I% ≈ 3.6% p.a., so R = 1 + 0.036/12 = 1.003. D forgot to divide by 12.	31% correct A 31% · B 15% · C 11% · D 33% · E 9%
☐	21	**B**	The scheduled repayment for 20 years at 3.14% is \$2246.53. Interest-only payments keep the balance at \$400 000, so solve for I% with N = 216, PV = 400 000 and PMT = −2246.53: about 2.21%.	33% correct A 9% · B 33% · C 25% · D 15% · E 16%
☐	22	**D**	$R_1 = 2 - 2 = 0$, then $R_2 = 2 - 0 = 2$.	44% correct A 5% · B 32% · C 13% · D 44% · E 6%
☐	23	**D**	True: effective equals nominal for annual compounding; it rises with more compounding periods; 12% p.a. nominal is 1% a month. False: an effective rate can't be lower than the nominal rate. Three.	33% correct A 3% · B 14% · C 45% · D 33% · E 4%
☐	24	**B**	Finance Solver: N = 8, I% = 4 × 0.52 = 2.08, PV = −10 500, FV = 12 700.95, quarterly: PMT ≈ −215.55.	32% correct A 7% · B 32% · C 31% · D 16% · E 13%
☐	25	**B**	Monday to Friday is four steps: $S_4 = T^4 S_0$ with $S_0 = (0.4, 0.6)$ gives Giorgio's ≈ 31.9%, about 32%. A (31%) and C (34%) used the wrong number of steps.	38% correct A 16% · B 38% · C 20% · D 13% · E 12%
☐	26	**E**	$A^T \times B$ is 4 × 2 times 2 × 3 = 4 × 3, so its transpose is 3 × 4. C^T is 4 × 3, and 4 × 3 times 3 × 4 is 4 × 4.	57% correct A 5% · B 9% · C 14% · D 14% · E 57%
☐	27	**E**	Columns add to 1: j = 0.4, m = 0.5 and l + n = 0.7. From S_1, 20m + 18 = 28 confirms m = 0.5 and 16 + 40n = 20 gives n = 0.1, so l = 0.6 = m + n.	41% correct A 19% · B 12% · C 11% · D 16% · E 41%
☐	28	**E**	B^T is 7 × 10, so $A(B^T)$ is 7 × 7 times 7 × 10: defined. AB and AB^2 don't match up, BA isn't square so it has no inverse, and $A^2 - BA$ subtracts matrices of different orders.	45% correct A 9% · B 15% · C 13% · D 17% · E 45%
☐	29	**A**	Find C first: $C = TS_0 - S_1 = (3, -1, 2)$. Then $S_2 = TS_1 - C = (23.04, 55.78, 16.18)$.	40% correct A 40% · B 12% · C 27% · D 14% · E 6%
☐	30	**D**	Tuesday's state is $TS_0 = (25, 35, 40)$. Stayers = 0.4 × 25 + 0.5 × 35 + 0.6 × 40 = 51.5%. C used Monday's state instead.	29% correct A 13% · B 17% · C 27% · D 29% · E 11%
☐	31	**A**	Row 1 of K has its 1 in column 3, so $p_{11} = m_{31}$. C (no movement) was a common guess.	42% correct A 42% · B 12% · C 25% · D 14% · E 7%

☐	32	**A**	Q (4 × 2) times N^T (2 × 3) gives each department's time for each person. Pre-multiplying by a 1 × 4 row of ones adds the departments: [1 1 1 1] × Q × N^T. B adds across the people instead.	31% correct A 31% · B 29% · C 7% · D 16% · E 16%
☐	33	**E**	People join only flavours (bipartite). The 8 vertices and 7 edges link up without a cycle, so it is connected, a tree and planar. All four.	11% correct A 4% · B 31% · C 32% · D 21% · E 11%
☐	34	**A**	Now: A–C–E–F–G–I = 25 weeks. If F runs weeks 10–13 and H follows (13–19), G ends at 17 and I starts at 19: 24 weeks, one less. The saving can't be the full three weeks because H now has to wait.	24% correct A 24% · B 36% · C 12% · D 20% · E 7%
☐	35	**C**	The graph is four vertices all joined to each other, which can be redrawn without crossings. Then Euler's formula gives 4 − 6 + f = 2, so f = 4. Most students counted the regions of the crossed drawing (5).	34% correct A 1% · B 2% · C 34% · D 61% · E 1%
☐	36	**D**	It is planar, has a cycle (a triangle), a bridge (the edge to the degree-1 vertex) and a Hamiltonian path. It has four odd vertices, so no Eulerian trail: four statements.	17% correct A 8% · B 31% · C 42% · D 17% · E 1%
☐	37	**C**	A zero means no direct connection. J–M, K–N and M–N are missing (two zeros each), and J, L, M and N have no loops (four diagonal zeros): 10. K has a loop, so its diagonal entry isn't zero.	33% correct A 9% · B 10% · C 33% · D 23% · E 24%
☐	38	**D**	Total each option: A 59.6, B 61.6, C 60.4, D 61.8, E 60.8 m. D is the largest.	39% correct A 8% · B 27% · C 18% · D 39% · E 6%
☐	39	**A**	None: the graph can be redrawn with no crossing edges, so it is already planar. Most students counted crossings instead.	21% correct A 21% · B 22% · C 37% · D 18% · E 2%
☐	40	**D**	The project takes 28 days with critical path B–E–G–H–I–K. The floats are A 4, C 4, D 5, F 3 and J 4, a total of 20.	34% correct A 8% · B 20% · C 23% · D 34% · E 13%

Multiple-choice score: ☐ **/ 40** **(state average 15.4)**

Exam 2 · written response

Question 1 (7 marks) · state average 3.3 / 7 (47%)

2020 VCAA Further Maths Exam 2 · Core · Question 5

a. Draw the least squares line on the scatterplot

☐ Straight line through two correctly calculated points across the data

Answer / notes: Calculate two points, e.g. waist 60 cm gives 1.104 and 130 cm gives 0.998, and join them with a ruler. Lines drawn by eye didn't score.

State average 0.4 / 1 (40%); 41% scored full marks. Mark distribution: 0: 59% · 1: 41%

Your mark: ☐ **/ 1**

b. Predicted body density for a 65 cm waist, to two decimal places

☐ 1.10 kg/L

Answer / notes: 1.195 − 0.001512 × 65 = 1.097 ≈ 1.10.

State average 0.7 / 1 (70%); 73% scored full marks. Mark distribution: 0: 27% · 1: 73%

Your mark: ☐ **/ 1**

c. Is that prediction extrapolation or interpolation?

- [ ] extrapolating

Answer / notes: Extrapolating: 65 cm is outside the range of the waist data, even though it is on the axes.

State average 0.4 / 1 (40%); 38% scored full marks. Mark distribution: 0: 62% · 1: 38%

Your mark: ____ **/ 1**

d. Interpret the slope

- [ ] On average, body density decreases by 0.001512 kg/L for each 1 cm increase in waist measurement

Answer / notes: Body density decreases by 0.001512 kg/L, on average, for each 1 cm increase in waist. “Decreases by −0.001512” or leaving out “each 1 cm” lost the mark.

State average 0.4 / 1 (40%); 44% scored full marks. Mark distribution: 0: 56% · 1: 44%

Your mark: ____ **/ 1**

e. Show that the residual for a 122 cm waist and density 0.995 is −0.02

- [ ] $0.995 - (1.195 - 0.001512 \times 122) = -0.0155 \approx -0.02$

Answer / notes: Predicted = 1.0105, and residual = actual − predicted = 0.995 − 1.0105 = −0.0155, which rounds to −0.02.

State average 0.5 / 1 (50%); 51% scored full marks. Mark distribution: 0: 49% · 1: 51%

Your mark: ____ **/ 1**

f. Correlation coefficient from $r^2 = 0.6783$, to three decimal places

- [ ] −0.824

Answer / notes: $r = -\sqrt{0.6783} = -0.824$. It is negative because the slope is. Most students missed the sign.

State average 0.3 / 1 (30%); 25% scored full marks. Mark distribution: 0: 75% · 1: 25%

Your mark: ____ **/ 1**

g. Does the residual plot support linearity? Explain

- [ ] Yes: the residuals show no clear pattern (randomly scattered)

Answer / notes: Yes: the residuals are scattered randomly with no clear pattern.

State average 0.6 / 1 (60%); 59% scored full marks. Mark distribution: 0: 41% · 1: 59%

Your mark: ____ **/ 1**

Question 2 (5 marks) · state average 2.2 / 5 (44%)

2020 VCAA Further Maths Exam 2 · Core · Question 6

a. Difference between the mean heights of women in their twenties and their eighties

- [ ] 10.4 cm

Answer / notes: 167.1 − 156.7 = 10.4 cm.

State average 0.9 / 1 (90%); 87% scored full marks. Mark distribution: 0: 13% · 1: 87%

Your mark: ____ **/ 1**

b. Describe the association: strength and direction

- [ ] strong, negative

Answer / notes: Strong and negative. Extra comments that contradicted this lost the mark.

State average 0.5 / 1 (50%); 51% scored full marks. Mark distribution: 0: 49% · 1: 51%

Your mark: ____ **/ 1**

c. Equation of the line through (20, 168) and (85, 157), to three significant figures

- [ ] mean height = 171 − 0.169 × mean age

Answer / notes: Slope = (157 − 168) ÷ (85 − 20) = −0.169; intercept = 168 + 0.169 × 20 ≈ 171. Many used CAS on the whole table instead of the two given points.

State average 0.2 / 1 (20%); 23% scored full marks. Mark distribution: 0: 77% · 1: 23%

Your mark: ☐ **/ 1**

d. Transform mean age to linearise the data and fit a least squares line, to four significant figures

- [ ] Uses a squared transformation of mean age
- [ ] mean height = 167.9 − 0.001621 × (mean age)²

Answer / notes: The data curve downwards, so square mean age. CAS on (age², height) gives mean height = 167.9 − 0.001621 × (mean age)². Few rounded to four significant figures correctly.

State average 0.6 / 2 (30%); 21% scored full marks. Mark distribution: 0: 62% · 1: 17% · 2: 21%

Your mark: ☐ **/ 2**

Question 3 (7 marks) · state average 3.9 / 7 (56%)

2021 VCAA Further Maths Exam 2 · Core · Question 1

a. Number of numerical variables in the table

- [ ] 3

Answer / notes: Three (high jump, shot-put and javelin); athlete number is a label. Many counted the 45 values instead.

State average 0.5 / 1 (50%); 52% scored full marks. Mark distribution: 0: 48% · 1: 52%

Your mark: ☐ **/ 1**

b. Mean high jump height

- [ ] 1.81 m

Answer / notes: 27.15 ÷ 15 = 1.81 m.

State average 0.9 / 1 (90%); 87% scored full marks. Mark distribution: 0: 13% · 1: 87%

Your mark: ☐ **/ 1**

c. Standardised score for a 14.50 m shot-put throw (mean 13.74, SD 1.43), to one decimal place

- [ ] 0.5

Answer / notes: z = (14.50 − 13.74) ÷ 1.43 ≈ 0.53, so 0.5.

State average 0.8 / 1 (80%); 78% scored full marks. Mark distribution: 0: 22% · 1: 78%

Your mark: ☐ **/ 1**

d. Percentage expected to jump higher than an athlete with z = −1.0

- [ ] 84%

Answer / notes: 16% lie below z = −1, so 84% jump higher. 16% was the common wrong answer.

State average 0.6 / 1 (60%); 60% scored full marks. Mark distribution: 0: 40% · 1: 60%

Your mark: ☐ **/ 1**

e. Why the high jump boxplot has no upper whisker

- [ ] The maximum (1.87) equals Q3

Answer / notes: Q3 and the maximum are both 1.87 m (four athletes jumped 1.87), so there is nothing to draw above the box.

State average 0.5 / 1 (50%); 47% scored full marks. Mark distribution: 0: 53% · 1: 47%

Your mark: ☐ **/ 1**

f. Minimum javelin throw that would show as an upper outlier

- [ ] Q1 = 40.88 and Q3 = 42.88, so upper fence = 42.88 + 1.5 × 2 = 45.88
- [ ] More than 45.88 m, so 45.89 m

Answer / notes: An outlier must be greater than the fence of 45.88 m, so the minimum is 45.89 m. Most gave 45.88.

State average 0.6 / 2 (30%); 9% scored full marks. Mark distribution: 0: 49% · 1: 42% · 2: 9%

Your mark: ☐ **/ 2**

Question 4 (5 marks) · state average 2.9 / 5 (58%)

2021 VCAA Further Maths Exam 2 · Core · Question 4

a. Draw the women's least squares line on the plot

- [ ] Straight line through two correctly calculated points, e.g. (1908, 75.3) and (2020, 48.0)

Answer / notes: Work out two points from 538.9 − 0.2430 × year, such as 75.256 in 1908 and 48.04 in 2020, and join them.

State average 0.5 / 1 (50%); 46% scored full marks. Mark distribution: 0: 54% · 1: 46%

Your mark: ☐ **/ 1**

b. Predicted women's minus men's winning time for 2024, to one decimal place

- [ ] Women 47.068 and men 44.394 (or equivalent working)
- [ ] 2.7 seconds

Answer / notes: (538.9 − 0.2430 × 2024) − (356.9 − 0.1544 × 2024) = 2.67 ≈ 2.7 s. Rounding too early was the main error.

State average 1.5 / 2 (75%); 67% scored full marks. Mark distribution: 0: 20% · 1: 12% · 2: 67%

Your mark: ☐ **/ 2**

c. First Olympic year the lines predict women will be faster than men

- [ ] Lines cross at about 2054.2
- [ ] 2056

Answer / notes: Solve 538.9 − 0.2430x = 356.9 − 0.1544x: x ≈ 2054.2. The next Olympic year after that is 2056.

State average 0.9 / 2 (45%); 40% scored full marks. Mark distribution: 0: 57% · 1: 4% · 2: 40%

Your mark: ☐ **/ 2**

Question 5 (2 marks) · state average 0.4 / 2 (20%)

2020 VCAA Further Maths Exam 2 · Core · Question 11

Number of further monthly repayments after the first seven years

- [ ] Finds the monthly repayment of $2400
- [ ] 150 more repayments

Answer / notes: Finance Solver from year 4 to year 7: N = 36, PV = 329 587.25, FV = −280 875.15 gives PMT = −2400. Then PV = 280 875.15, PMT = −2400, FV = 0 gives N ≈ 149.7, so 150 more (the last one smaller). Entering PV and FV with the same sign was a common error.

State average 0.4 / 2 (20%); 19% scored full marks. Mark distribution: 0: 77% · 1: 3% · 2: 19%

Your mark: ☐ **/ 2**

Question 6 (3 marks) · state average 1.2 / 3 (40%)

2021 VCAA Further Maths Exam 2 · Core · Question 8

a. Balance after the first repayment

- [ ] $569 377

Answer / notes: 1.001 × 570 000 − 1193 = $569 377.

State average 0.7 / 1 (70%); 68% scored full marks. Mark distribution: 0: 32% · 1: 68%

Your mark: ☐ **/ 1**

b. Show the interest rate is 2.6% p.a.

- [ ] $(1.001 - 1) \times 26 \times 100 = 2.6$

Answer / notes: 0.1% a fortnight × 26 fortnights = 2.6% p.a. CAS syntax doesn't count as showing.

State average 0.3 / 1 (30%); 27% scored full marks. Mark distribution: 0: 73% · 1: 27%

Your mark: ____ **/ 1**

c. Final (larger) repayment that clears the loan, to the nearest cent

- [ ] $1198.59

Answer / notes: Finance Solver gives N ≈ 650.005, so 650 payments with the last one larger. The balance after 649 payments, plus a fortnight's interest, is $1198.59. $1197.39 left out that interest.

State average 0.2 / 1 (20%); 15% scored full marks. Mark distribution: 0: 85% · 1: 15%

Your mark: ____ **/ 1**

Question 7 (3 marks) · state average 0.8 / 3 (27%)

2021 VCAA Further Maths Exam 2 · Core · Question 9

a. Value after two months, showing recursive calculations

- [ ] Factor 1.00425
- [ ] $V_1 = 152\,178.83$ and $V_2 = 151\,925.59$

Answer / notes: $V_1 = 1.00425 \times 152\,431 - 900 = 152\,178.83$; $V_2 = 1.00425 \times 152\,178.83 - 900 = \$151\,925.59$. Common errors: adding the 900, or not showing both steps.

State average 0.7 / 2 (35%); 21% scored full marks. Mark distribution: 0: 58% · 1: 21% · 2: 21%

Your mark: ____ **/ 2**

b. One-off amount to add after two years so the payments last the same time at 4.6% p.a.

- [ ] $7039.20

Answer / notes: After 24 payments the balance is $146 073.74. At 4.6%, 276 more payments of $900 need $153 112.94. The difference is $7039.20. Only 10% of marks were earned here.

State average 0.1 / 1 (10%); 8% scored full marks. Mark distribution: 0: 92% · 1: 8%

Your mark: ____ **/ 1**

Question 8 (4 marks) · state average 1.5 / 4 (38%)

2022 VCAA Further Maths Exam 2 · Core · Question 8

a. Balance after two months, showing recursive calculations

- [ ] $L_1 = 578\,114.74$ and $L_2 = 576\,225.71$

Answer / notes: $L_1 = 1.002 \times 580\,000 - 3045.26 = 578\,114.74$; $L_2 = 1.002 \times 578\,114.74 - 3045.26 = \$576\,225.71$.

State average 0.5 / 1 (50%); 44% scored full marks. Mark distribution: 0: 56% · 1: 44%

Your mark: ____ **/ 1**

b. Annual interest rate

- [ ] 2.4%

Answer / notes: 0.2% a month × 12 = 2.4% p.a. 0.2% was a common answer.

State average 0.5 / 1 (50%); 51% scored full marks. Mark distribution: 0: 49% · 1: 51%

Your mark: ____ **/ 1**

c. Amount by which the final repayment is smaller, to the nearest cent

☐ $0.15

Answer / notes: Finance Solver: N ≈ 239.99, so 240 payments. With N = 240 the balance overshoots by FV ≈ 0.15, so the last payment is $0.15 smaller.

State average 0.3 / 1 (30%); 23% scored full marks. Mark distribution: 0: 77% · 1: 23%

Your mark: ☐ **/ 1**

d. Multiplying factor that would repay the loan one year sooner, to four decimal places

☐ 1.0016

Answer / notes: Solve for I% with N = 228, PV = 580 000, PMT = −3045.26, FV = 0: about 1.9466% p.a. The factor is 1 + 0.019466/12 ≈ 1.0016.

State average 0.2 / 1 (20%); 19% scored full marks. Mark distribution: 0: 81% · 1: 19%

Your mark: ☐ **/ 1**

Question 9 (5 marks) · state average 1.7 / 5 (34%)

2021 VCAA Further Maths Exam 2 · Matrices module · Question 3

a. How many shoppers bought a different brand in August from July

☐ 1160

Answer / notes: Stayers = 0.85 × 3200 + 0.80 × 2000 + 0.90 × 2800 = 6840, so 8000 − 6840 = 1160 switched.

State average 0.3 / 1 (30%); 28% scored full marks. Mark distribution: 0: 72% · 1: 28%

Your mark: ☐ **/ 1**

b. Complete S_1

☐ S_1 = (3060, 1900, 3040)

Answer / notes: TS_0 = (3060, 1900, 3040).

State average 0.8 / 1 (80%); 78% scored full marks. Mark distribution: 0: 22% · 1: 78%

Your mark: ☐ **/ 1**

c. Percentage of August's Carmani buyers who also bought Carmani in July

☐ 89%

Answer / notes: 0.85 × 3200 = 2720 stayed, out of 3060 August buyers: 2720 ÷ 3060 ≈ 89%. 85% came straight from the matrix and missed the point.

State average 0.2 / 1 (20%); 17% scored full marks. Mark distribution: 0: 83% · 1: 17%

Your mark: ☐ **/ 1**

d. Calculation showing Ohana is bought by 50% of shoppers in the long run

☐ $T^n \times S_0$ for large n (e.g. n = 50) gives O = 4000, which is half of 8000

Answer / notes: Raise T to a large power and multiply by S_0: Ohana settles at 4000 of the 8000 shoppers.

State average 0.2 / 1 (20%); 22% scored full marks. Mark distribution: 0: 78% · 1: 22%

Your mark: ☐ **/ 1**

e. Value of k in $R_{n+1} = TR_n + B$, given R_2

☐ 200

Answer / notes: $R_1 = TR_0 + B$ and $R_2 = TR_1 + B$. R_1 = (3260, 2000, 3040 + k), and the Ohana element of R_2 works out to 3262 + 1.9k = 3642, so k = 200.

State average 0.2 / 1 (20%); 15% scored full marks. Mark distribution: 0: 85% · 1: 15%

Your mark: ☐ **/ 1**

Question 10 (2 marks) · state average 0.4 / 2 (20%)

2021 VCAA Further Maths Exam 2 · Matrices module · Question 4

Complete the results matrix using all the information

- [ ] Rows: I 0 1 1 1 0; J 0 0 1 1 1; K 0 0 0 1 0
- [ ] L 0 0 0 0 1; M 1 0 1 0 0

Answer / notes: Katie's row is given (she beat only Leslie). Ike and Joelene each won 3, Mikki 2 and Leslie 1, and each pair has exactly one 1. The two-step totals (5, 4, 1, 2, 4) then fix the rest: Ike beat Joelene, Katie and Leslie; Joelene beat Katie, Leslie and Mikki; Leslie beat Mikki; Mikki beat Ike and Katie.

State average 0.4 / 2 (20%); 10% scored full marks. Mark distribution: 0: 70% · 1: 20% · 2: 10%

Your mark: ☐ **/ 2**

Question 11 (3 marks) · state average 1.1 / 3 (37%)

2022 VCAA Further Maths Exam 2 · Matrices module · Question 4

a. Number planning to vote for Ali after one day

- [ ] 146

Answer / notes: Fill in the missing percentages so each candidate's shares add to 100%: 20% of Lee's and 30% of Max's supporters move to Ali. 0.4 × 160 + 0.2 × 140 + 0.3 × 180 = 146.

State average 0.3 / 1 (30%); 32% scored full marks. Mark distribution: 0: 68% · 1: 32%

Your mark: ☐ **/ 1**

b. Who becomes captain after Max withdraws, and with how many votes

- [ ] Uses $T^6 \times S_1$ with $S_1 = (146, 154, 180)$
- [ ] Ali with 280 votes

Answer / notes: $S_1 = (146, 154, 180)$. Six more days with the new matrix: $T^6S_1 \approx (280, 200, 0)$. Ali wins with 280 votes.

State average 0.8 / 2 (40%); 39% scored full marks. Mark distribution: 0: 60% · 1: 1% · 2: 39%

Your mark: ☐ **/ 2**

Question 12 (4 marks) · state average 2.1 / 4 (53%)

2020 VCAA Further Maths Exam 2 · Networks module · Question 3

a. Shortest distance from S to O

- [ ] 3.2 km

Answer / notes: S–T–U–V–O = 0.6 + 1.2 + 0.6 + 0.8 = 3.2 km. 3.3 km was a common slip.

State average 0.6 / 1 (60%); 58% scored full marks. Mark distribution: 0: 42% · 1: 58%

Your mark: ☐ **/ 1**

b.i. Name for a route along every track exactly once

- [ ] Eulerian trail

Answer / notes: An Eulerian trail (not a path).

State average 0.7 / 1 (70%); 66% scored full marks. Mark distribution: 0: 34% · 1: 66%

Your mark: ☐ **/ 1**

b.ii. Station where that route finishes

- [ ] P

Answer / notes: S and P are the only odd-degree vertices, so a trail starting at S ends at P.

State average 0.6 / 1 (60%); 60% scored full marks. Mark distribution: 0: 40% · 1: 60%

Your mark: ☐ **/ 1**

c. Track that must be repeated to visit every station and return to S in the minimum distance

☐ S and T

Answer / notes: T, M and R each have only two tracks, and all three join S, so a circuit through every station would need all three of S's tracks. One must be run out and back, and the shortest is S–T (0.6 km).

State average 0.2 / 1 (20%); 25% scored full marks. Mark distribution: 0: 75% · 1: 25%

Your mark: ☐ **/ 1**

Question 13 (4 marks) · state average 1.1 / 4 (28%)

2020 VCAA Further Maths Exam 2 · Networks module · Question 5

a. Where the dummy activity goes

☐ From the end of B to the start of C

Answer / notes: C needs A and B but F needs only B, so B ends at its own vertex where F starts, with a dummy to C.

State average 0.3 / 1 (30%); 26% scored full marks. Mark distribution: 0: 74% · 1: 26%

Your mark: ☐ **/ 1**

b. Duration of activity C

☐ 2

Answer / notes: C starts at 5 and D and E start at 7, so C takes 2 months.

State average 0.5 / 1 (50%); 49% scored full marks. Mark distribution: 0: 51% · 1: 49%

Your mark: ☐ **/ 1**

c. The four activities that have float

☐ A, E, F, H

Answer / notes: The critical path is B–C–D–G–I (20 months). The non-critical activities are A, E, F and H.

State average 0.2 / 1 (20%); 23% scored full marks. Mark distribution: 0: 77% · 1: 23%

Your mark: ☐ **/ 1**

d. Minimum completion time if one activity is crashed

☐ 17

Answer / notes: Reducing B shortens every path through it. Cut B by 3 and B–C–D–G–I falls to 17, while A–C–D–G–I is also 17, so 17 months. Crashing D or G only reaches 19 because of B–F–H–I.

State average 0.1 / 1 (10%); 12% scored full marks. Mark distribution: 0: 88% · 1: 12%

Your mark: ☐ **/ 1**

Question 14 (4 marks) · state average 1.5 / 4 (38%)

2022 VCAA Further Maths Exam 2 · Networks module · Question 3

a. Capacity of Cut 1

☐ 74

Answer / notes: Add only the pipes that cross from the source side to the sink side: 15 + 15 + 14 + 11 + 19 = 74. The pipe from M to L runs the other way, so it doesn't count.

State average 0.6 / 1 (60%); 62% scored full marks. Mark distribution: 0: 38% · 1: 62%

Your mark: ☐ **/ 1**

b. Maximum flow from source to sink

☐ 44

Answer / notes: Find the minimum cut: 44 L/min. Many different wrong approaches were seen.

State average 0.4 / 1 (40%); 37% scored full marks. Mark distribution: 0: 63% · 1: 37%

Your mark: ☐ **/ 1**

c. Pipe to reverse for the largest increase in flow

- [ ] From Q to P

Answer / notes: Q→P feeds P, which has no way out to the sink. Reversed, it carries flow from N and O through P and Q to the sink: a 6 L/min gain.

State average 0.5 / 1 (50%); 44% scored full marks. Mark distribution: 0: 56% · 1: 44%

Your mark: ____ **/ 1**

d. Pipe whose capacity should be increased for the largest gain, and its new capacity

- [ ] L to R
- [ ] At least 23

Answer / notes: L→R is in the minimum cut, and more flow can reach L through J and K. Examiners' answer: increase L→R to at least 23 L/min. Almost nobody got this.

State average 0.0 / 1 (2%); 2% scored full marks. Mark distribution: 0: 98% · 1: 2%

Your mark: ____ **/ 1**

Exam 1: ____ **/ 40** **Exam 2:** ____ **/ 58** **Total:** ____ **/ 98**

State average on these exact questions: Exam 1 15.4 / 40, Exam 2 24.1 / 58, total 39.5 / 98 (40%). The notes paraphrase the VCAA examiners' reports; read the full reports for sample answers.